

Common Attacks and Their Effectiveness

Historical cyber incidents like the **LoveLetter virus (ILOVEYOU)** and the **Morris worm** heavily impacted early security models, leading directly to the creation of **CSIRTs** (Computer Security Incident Response Teams). Today's threat actors build upon those concepts with refined, modern variations of phishing, malware, and social engineering.

■ Phishing Attacks

Phishing uses digital communication channels like emails, SMS, or voice calls to deceive individuals into handing over confidential credentials or launching malicious files.

- **Business Email Compromise (BEC):** Threat actors spoof or compromise high-level corporate email addresses to make legitimate-looking operational requests, aiming for financial gain or theft of information.
- **Spear Phishing:** Highly personalized email attacks targeting specific users or groups. The attacker researches targets ahead of time so messages appear to come from trusted internal or external contacts.
- **Whaling:** A focused sub-category of spear phishing explicitly targeting executive leadership (CEOs, CFOs) to extract maximum financial advantages or highly critical corporate intelligence.
- **Vishing:** Voice phishing carried out via electronic telephone systems to exploit trust, impersonate official entities, and trick individuals into supplying private authentication numbers.
- **Smishing:** Text message-based phishing attacks built to trick mobile phone users into disclosing account details or clicking malicious links while on the move.

■ Malware Exploit Types

Malware includes harmful software components built intentionally to compromise hardware nodes, map networks, or secure an illicit data advantage against private entities or state territories.

- **Viruses:** Malicious payloads attached directly to clean files that rely entirely on human interaction to run and replicate. Once opened, they write dark code into local software applications.
- **Worms:** High-risk self-replicating programs that spread across server infrastructure automatically without any human input, rapidly consuming available network bandwidth.
- **Ransomware:** Highly disruptive malicious code that systematically encrypts a victim company's databases, holding administrative control hostage until financial extortion demands are paid.
- **Spyware:** Invisible monitoring software deployed to systematically record keystrokes, extract communications, track geographic paths, and gather user profiles without explicit legal consent.

■ Social Engineering & Core Principles

Social engineering exploits common human behavior patterns to trick people into giving away access credentials. Because humans are naturally cooperative, threat actors use seven psychological principles to establish false credibility:

- **Social Media Phishing:** Reviewing open profiles online to collect hyper-targeted data before sending a malicious communication.
- **Watering Hole Attack:** Modifying a specific legitimate website regularly frequented by the target group to silently distribute a drive-by malware payload.
- **USB Baiting:** Leaving a virus-loaded flash drive in an administrative area or parking lot to exploit an employee's natural curiosity.
- **Physical Social Engineering:** Sneaking directly into secure buildings by dressing as vendors or delivery drivers to bypass electronic front doors.
- **Authority:** Impersonating executives or law enforcement officers because individuals are conditioned to obey official figures.
- **Intimidation:** Employing psychological bullying or immediate organizational threats to bypass standard vetting procedures.
- **Consensus / Social Proof:** Claiming that everyone else in the workspace has already performed the requested task to normalise the bad action.
- **Scarcity:** Presenting assets or technical help as extremely limited resources to force immediate, reckless engagement.
- **Familiarity:** Creating common ground by highlighting shared professional backgrounds or mutual associates to lower user alertness.
- **Trust:** Establishing an ongoing, long-term personal connection to harvest data incrementally over time.
- **Urgency:** Forcing tight artificial deadlines so victims act impulsively without verifying the communication.